

OPERATING SYSTEMS

Storage Management

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OPERATING SYSTEMS

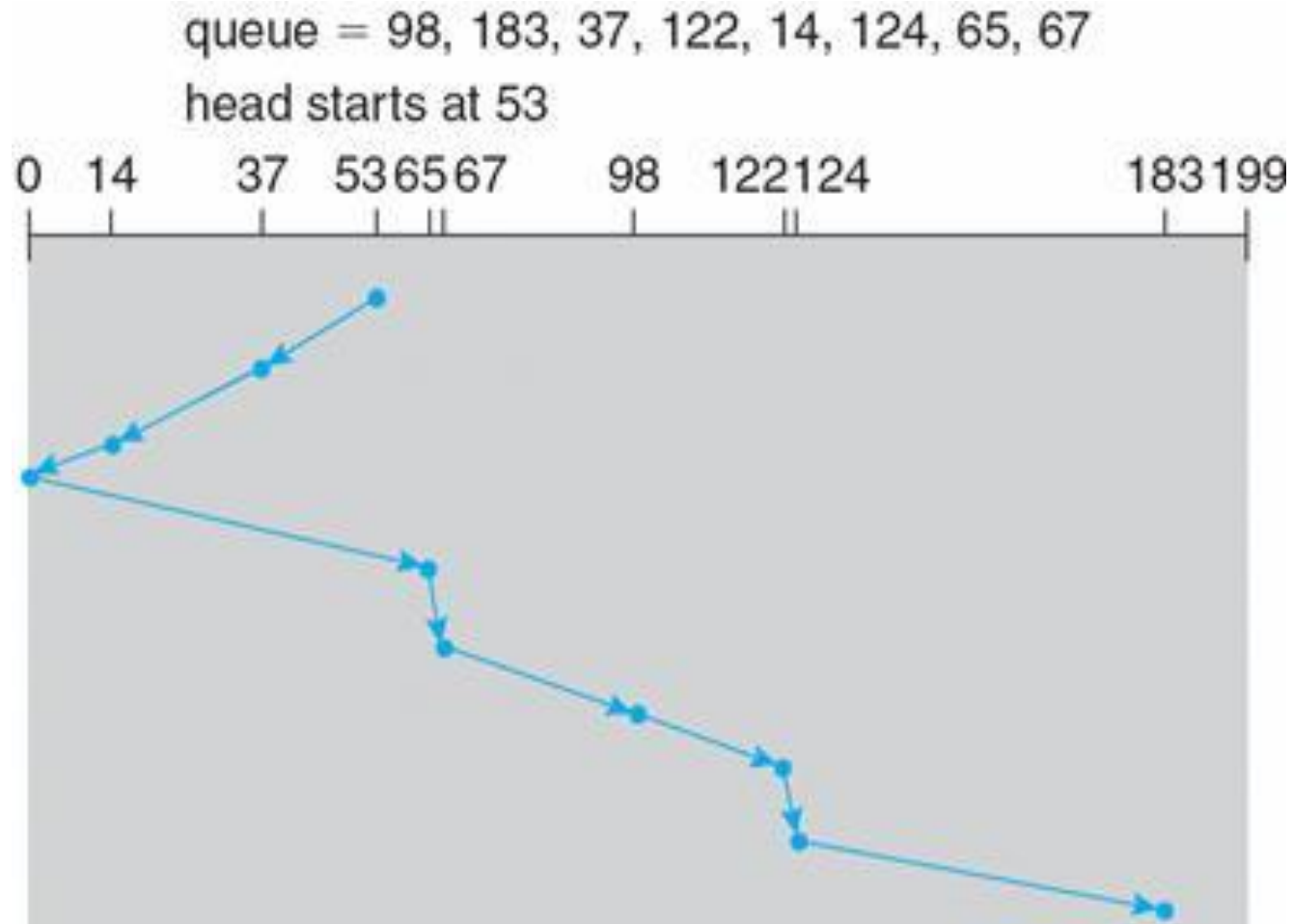
Disk Scheduling Algorithm - SCAN

The disk arm **starts at one end** of the disk, and moves toward the other end, servicing requests until it gets to the other end of the disk, where the head movement is **reversed** and servicing continues.

It is also called elevator algorithm

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Disk Scheduling Algorithm - SCAN



Head movement shows 236 cylinders

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Disk Scheduling Algorithm - SCAN

$$\begin{aligned}\text{Total Head movement} &= |53-37| + |37-14| + \\ &\quad |14-0| + |0-65| + \\ &\quad |65-67| + |67-98| + |98-122| + \\ &\quad |122-124| + |124-183| \\ &= 16 + 23 + 14 + 65 + 2 + 31 + 24 + 2 + 59 \\ &= 236\end{aligned}$$

Head movement shows 236 cylinders

Advantages

- High throughput
- Low variance of response time
- Average response time

Disadvantages

- Long waiting time for requests for locations just visited by disk arm

Provides a more uniform wait time than SCAN

The head moves from one end of the disk to the other, servicing requests as it goes

When it reaches the other end, however, it immediately **returns to the beginning of the disk**, without servicing any requests on the return trip.

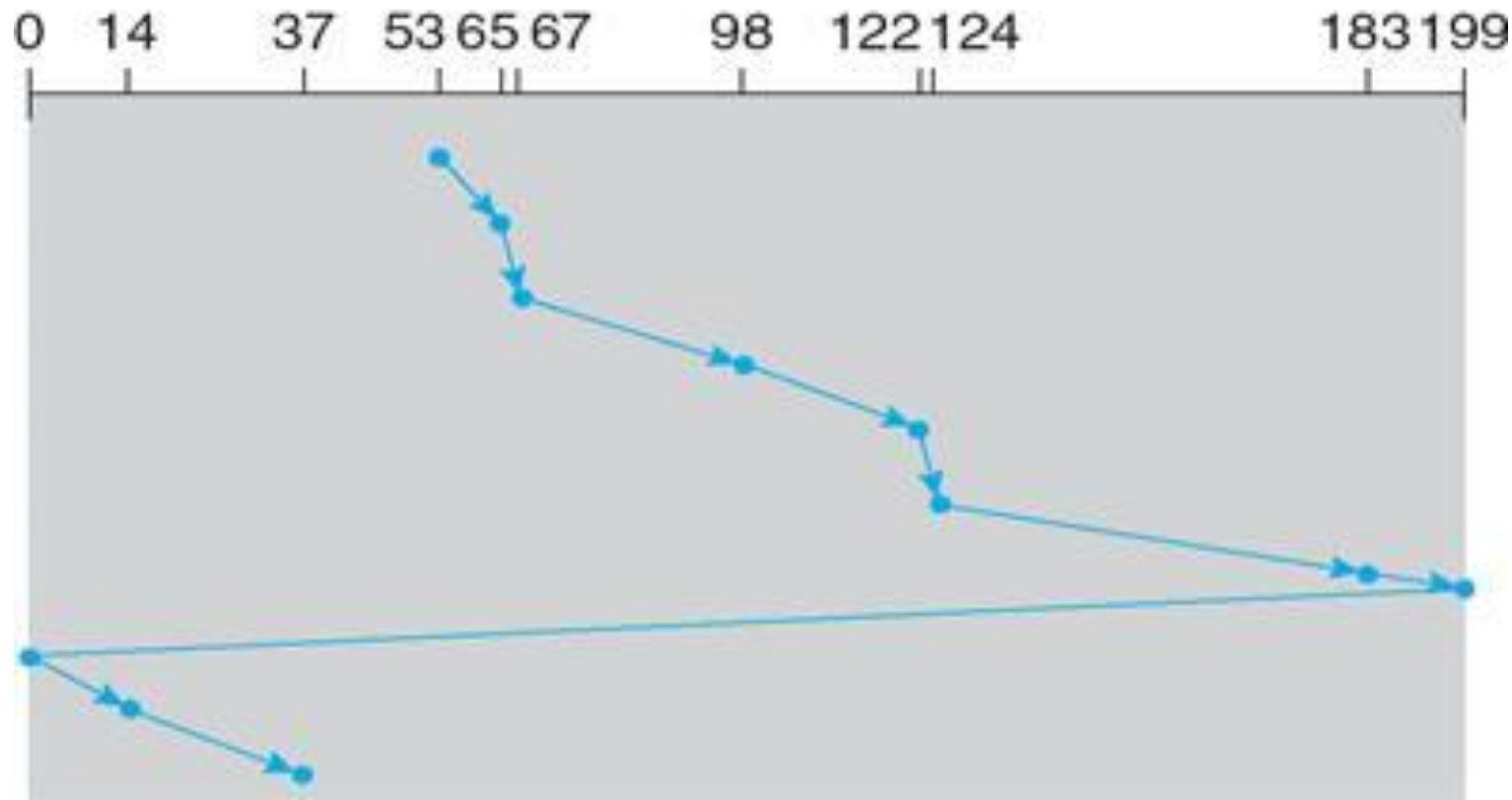
Treats the cylinders as a **Circular** list that wraps around from the last cylinder to the first one

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Disk Scheduling Algorithm - CSCAN

queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



Head movement shows 382 cylinders

$$\begin{aligned}\text{Total Head movement} &= |53-65| + |65-67| + \\ &\quad |67-98| + |98-122| + \\ &\quad |122-124| + |124-183| + \\ &\quad |183-199| + |199-0| + \\ &\quad |0-14| + |14-37| \\ &= 12 + 2 + 31 + 24 + 2 + 59 + 16 + 199 + 14 + 23 \\ &= 382\end{aligned}$$

Head movement shows 382 cylinders

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Disk Scheduling Algorithm - CSCAN

Advantages

- Provides more uniform wait time compared to SCAN



THANK YOU

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